

The Complete Option Pricing with Reinforcement Learning From A-Z in Python

Introduction to American Options

- American options are a type of financial derivative that give the holder the right, **but not the obligation**, to buy or sell an underlying asset at a specified price on or before a certain date.
- **Call Option:** Gives the holder the **right to buy the underlying asset at the strike price.**
- **Put Option:** Gives the holder the **right to sell the underlying asset at the strike price.**

Example: American Put Option

Option Type: American Put Option

Strike Price (K): \$100

Underlying Asset Price (S): \$95 initially

Expiration Date: 3 days from now

Premium (Initial Price of Option): \$5

Price of Underlying Asset over the next 3 days:

Day	Asset Price (S)
0	\$95
1	\$92
2	\$88
3	\$85

Example: American Put Option

Example Day 1:

Asset Price (S_1): \$92

Exercise Value: $\max(K - S_1, 0) = \max(100 - 92, 0) = \8

If exercised:

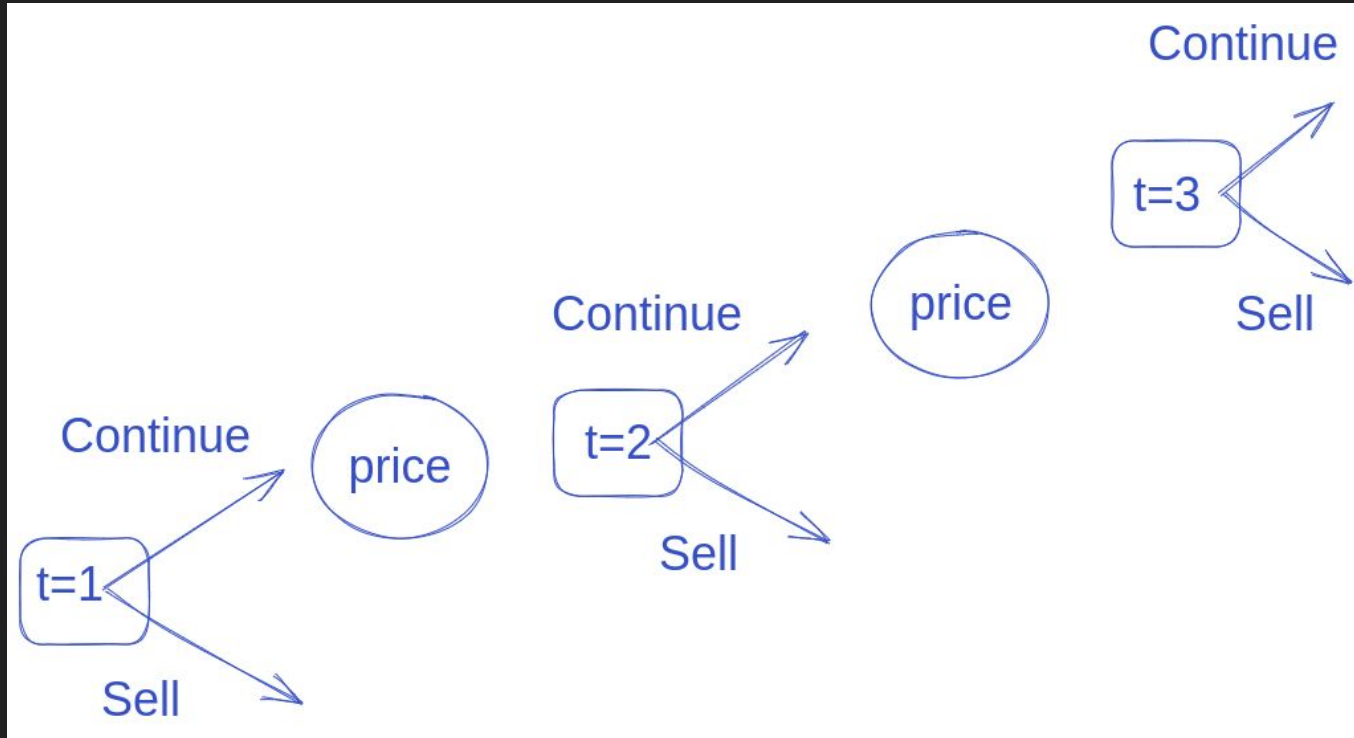
Sell the asset at \$100 (strike price)

Buy back at market price: \$92

Profit: \$8 - \$5 (premium) = \$3

Day	Asset Price (S)
0	\$95
1	\$92
2	\$88
3	\$85

American Options



Profit in American Options



American Option VS European Option

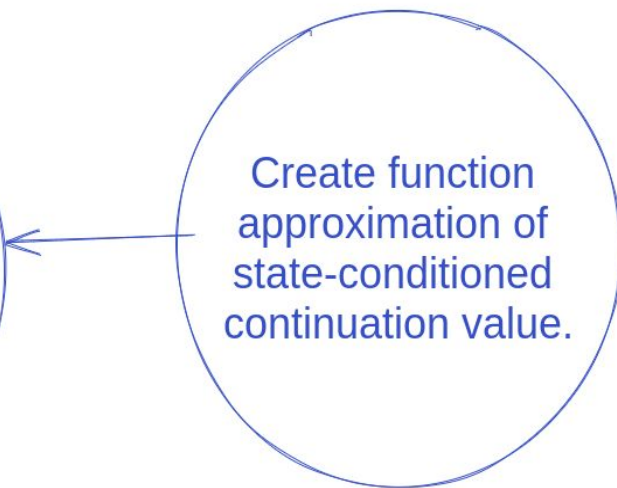
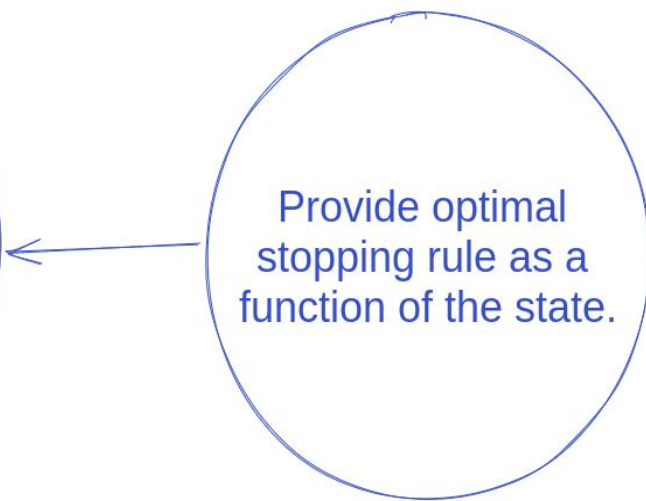
- Unlike European options, which can only be exercised at maturity, American options can be exercised at any point prior to expiration.
- This added optionality complicates the valuation process, as the holder must consider the optimal (early) exercise strategy to maximize their return.

Optimal Exercise of American Options

Valuation of the option

When to stop?

Function Approximation



Valuation of American Option:

- A) The value of Option with Reinforcement Learning based method (RL).
(American_Option_RL.ipynb)
- B) The value of Option with Least-Squares Monte Carlo method (LSM).
(American_Option_LSM.ipynb)
- C) The value of Option with Binomial - Lattice method (Bin-Tree)
(American_Option_BOPM.ipynb)

The Value of Option with Reinforcement Learning

